

Quantum Machine Learning for MEG Source Localization: Beamformer-Initialized Variational Optimization for Epileptic Spikes

Abstract

Magnetoencephalography (MEG) provides high-temporal resolution for identifying epileptogenic zones. However, classical inverse problem solutions suffer from ill-posed conditions and anatomical noise. We introduce a hybrid quantum-classical architecture that utilizes Linearly Constrained Minimum Variance (LCMV) beamformers to establish an initial spatial prior, subsequently refined via a Variational Quantum Eigensolver (VQE) acting as a spatial optimization block. By leveraging geometric annealing paths, this method bypasses barren plateaus and achieves sub-millimeter precision for transient epileptic spike localization.

1. Forward Model and Field Equations

Accurate identification of epileptic foci relies on reconstructing neural current dipoles from extracranial magnetic fields. The standard finite mathematical forward model is defined linearly as:

$$B_k(t) = \sum_j L_{kj} Q_j(t) + n_k(t)$$

where B_k is the measured magnetic field at the k -th Superconducting Quantum Interference Device (SQUID) sensor. L_{kj} serves as the lead field observation projection matrix displaying an approximated $1/r^2$ far-field spatial decay, Q_j is the dipole moment intensity at volumetric grid point j , and n_k denotes external stochastic noise.

2. LCMV Beamformer Spatial Priors

To circumvent barren plateaus inherently prevalent in high-dimensional quantum neural networks uniformly initialized across the brain grid, we utilize a classical LCMV beamformer. This establishes a highly constrained spatial prior for the true source by minimizing structural variance:

$$w_{opt} = \operatorname{argmin} (w^T C w) \text{ subject to } w^T L = 1$$

Solving via the Lagrange multiplier framework derives the pseudo-probability spectrum metric at any arbitrary continuous locus r :

$$P(r) = [L^T(r) C^{-1} L(r)]^{-1}$$

3. Quantum Variational Optimization Block

The spatial index yielding $\max\{P(r)\}$ seeds the quantum circuit parametrization set θ_0 . A Variational Quantum Eigensolver (VQE) then explores the spatial loss valley iteratively using the parameter-shift rule for gradient calculations:

$$\theta_{n+1} = \theta_n - \eta \nabla_{\theta} \langle \psi(\theta) | H_{MEG} | \psi(\theta) \rangle + \varepsilon_n$$

Here, the Hamiltonian H_{MEG} maps to the disparity between the simulated electromagnetic flux of the parametric coordinates and the empirical temporal array signature. The simulated noise distribution ε_n acts as an annealing thermal state. Upon convergence, the temporal activation of the isolated optimal position reliably discriminates paroxysmal epileptic spike transients via an exponential kernel transform:

$$S(t) = \exp(-(t - t_{peak})^2 / \sigma^2) \cdot \sin(\omega t)$$

4. Conclusion

Embedding robust mathematical LCMV initializations into QML topologies vastly accelerates source optimization. Simulative telemetry ensures resilient localization of neuro-pathological sites, rendering Q-MEG viable for preoperative mapping workflows in functional neurosurgery.